

Name: _____

Solve's

Date: _____



Maths Year 9

Investigation 2, 2011

Topic – Binary numbers

 24

_____ %

Total Time: 30 minutes

Weighting: _____ % of the year.

Equipment: Formula sheets, Scientific calculator

Penalties (only deducted once in the whole assessment):

- ☐ Rounding - Incorrect/inappropriate (-1)
- ☐ Units - Incorrect/missing (-1)

Important Information:

Although the take-home component is not worth any marks, it is essential in preparation for the in-class component. Knowledge and skills gained will be extended in the in-class validation component. This in-class validation will be completed under test conditions on the day in which this take-home component is due. Answers should be rounded appropriately. All working should be shown in the space provided. Solutions without working may not be awarded full marks. Please take the marks for each question into account when answering the question.

Contact will be made to parent(s) if the take-home component is not available for submission (at the start of the lesson).

Date out: Week _____ Date _____ Date Due: Week _____ Date _____

Take home component weighting: 0% of the year

In-class component weighting: _____ % of the year.

AIM:

In this assessment, you will be investigating binary numbers.

1. [3 marks:]

Complete the following table

Indice	2^0	2^1	2^2	2^3	2^4	2^5	2^6	2^7	2^8
Value	1	2	4	8	16	32	64	128	256

-1 each mistake.

2. [8 marks 5,3]

a) 1101_2

Digit	1	1	0	1
Place value	8	4	2	1
Place value as an indice	2^3	2^2	2^1	2^0

$$\begin{aligned}
 1101_2 &= 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \quad (\text{expanded notation index form}) \\
 &= 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 \quad (\text{expanded notation}) \\
 &= 13
 \end{aligned}$$

So 1101_2 is the number 13

b) 10100_2

$$\begin{aligned}
 &1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0 \quad \checkmark \\
 &= 16 + 4 \quad \checkmark \\
 &= 20 \quad \checkmark
 \end{aligned}$$

3. [6 marks:]

Convert the following numbers to binary

a) 12

$$\begin{aligned}
 12 &= 8 + 4 \\
 &= 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0 \quad \checkmark \\
 &= 1100_2 \quad \checkmark
 \end{aligned}$$

b) 45

$$\begin{aligned}
 45 &= 32 + 8 + 4 + 1 \\
 &= 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \quad \checkmark \\
 &= 101101_2 \quad \checkmark
 \end{aligned}$$

c) 82

$$\begin{aligned}
 82 &= 64 + 16 + 2 \\
 &= 1 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 \quad \checkmark \\
 &= 1010010_2 \quad \checkmark
 \end{aligned}$$

4. [2 marks:]

Sally is completing a binary project and is looking at the number 11031 on her page. She can't remember if it was a binary number or a normal (base ten) number. Is it possible to tell? Explain your answer

It has a 3 ✓ binary only uses 1 and 0's ✓

5. [2 marks:]

Max lives on the ground floor of an apartment building in number 52. His friend posts Max a letter and decided to write Max's number in binary. Which door should Max knock on to retrieve his letter?

$$\begin{aligned} 52 &= 32 + 16 + 4 \\ &= 2^5 + 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0 \quad \checkmark \\ &= 110100 \quad \checkmark \end{aligned}$$

6. [3 marks:]

Eric messed up on his assignment and used base-3 instead of binary, he wrote 101_3 . Use your knowledge of base-2 (binary) and relate it to base-3.

What is this number in

a) Base ten (normal)

$$\begin{aligned} &1 \times 3^2 + 0 \times 3^1 + 1 \times 3^0 \quad \checkmark \\ &= 10 \quad \checkmark \end{aligned}$$

b) Binary

$$\begin{aligned} &8 + 2 \\ &1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 \\ &1010_2 \end{aligned}$$